SRL

Synthetic accuracy: exact-nonlinear holdout

Task. Recover differential vascular conductivity $\Delta\sigma$ from 32 boundary voltages. Metrics compare against known ground truth $\Delta\sigma^*$: 32 exact-nonlinear holdout phantoms, never seen in training.

Method	Image corr. ↑	Dice ↑	Image RMSE ↓	BG RMS ↓
Newton, one step (PVI-style)	0.265	0.295	0.00506	0.00063
Iterative LM, 2 steps (no GCN)	0.291	0.394	0.00503	**0.00059**
Selected GCNM ($\alpha = 1, \beta = 0.25$)	**0.557**	**0.485**	**0.00432**	0.00128

Takeaway. Iterating the physics alone barely moves correlation (0.265→0.291). The jump to 0.557 is what the **learned** prior adds. The selected model localizes vessels far better (Dice 0.485) at a modest cost in background cleanliness.

Metric key: corr. = spatial pattern match versus truth; Dice = vessel-location overlap (0–1); BG RMS = false signal in non-vessel regions. Bold = best value in each metric column.

Source: technical report Tables 8 and 10; common 32-phantom exact-nonlinear holdout.